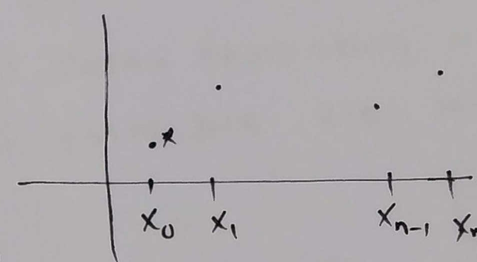
Numerical Differentiation

In this section, we will learn how to approximate from the data. Consider the following data of $(n+1)$ points.

$$x_0, \; x_1, \; x_2, \; x_3 \; \cdots \; x_{n-1}, \; x_n$$

$$f(x_0) \; f(x_1), \; f(x_2), \; f(x_3), \; \cdots \; f(x_{n-1}), \; f(x_n)$$



Since we don't have function form of f, we will not be able to find derivative in the continuous sense. One approach is to approximate the data by Lagrange polynomial and then differentiate that polynomial.

Lagrange polynomial for $(n+1)$ data points is given by

$$f(x) = \sum_{k=0}^{n} L_k(x) f(x_k) + \frac{f^{(n+1)}(\xi(x))}{(n+1)!}(x-x_0)(x-x_1)\cdots(x-x_n). \quad \text{—} (1)$$

Differentiating (1), we arrive at

$$f'(x) = \sum_{k=0}^{n} L'_k(x) f(x_k) + \frac{d}{dx}\left[\frac{f^{(n+1)}(\xi(x))}{(n+1)!}\right](x-x_0)(x-x_1)\cdots(x-x_n)$$

$$+ \frac{f^{(n+1)}(\xi(x))}{(n+1)!}\frac{d}{dx}\left[(x-x_0)(x-x_1)\cdots(x-x_n)\right]. \quad \text{—} (2)$$

Now, if we evaluate f' at any x_j given in the data, this further simplifies eq (2)

$$f'(x_j) = \sum_{k=0}^{n} L_k'(x_j) f(x_k) + \frac{f^{(n+1)}(\xi(x_j))}{(n+1)!} \prod_{\substack{k=0 \\ j\neq k}}^{n} (x_j - x_k) \quad ②$$

This called (n+1) point formula.

③

Lets say we are interested in finding derivative at $x=x_0$, i.e, $f'(x_0)$. We don't need to consider entire data for this purpose. Becouse derivative is a local quantity and requires information in close vicinity of the point only. Therefore, we may chouse to select few points close to x_0. Base on that we make several cases depending on how many points we take around the point we are interested in finding the derivative at.

① Consider only two points x_0, x_1 and ~~substitu~~. write ③ ~~for this data~~ to find derivative at x_0

$$f'(x_0) = L_0'(x_0) f(x_0) + L_1'(x_0) f(x_1) + \frac{f^{(2)}(\xi(x_0))}{2!}(x_0 - x_1)$$

$$f'(x_0) = \frac{d}{dx}\left[\frac{x-x_1}{x_0-x_1}\right] f(x_0) + \frac{d}{dx}\left[\frac{x-x_0}{x_1-x_0}\right] f(x_1) + \frac{f^{(2)}(\xi_0)}{2!}(x_0-x_1)$$

$\because \xi_0 = \xi(x_0)$

$$f'(x_0) = \frac{1}{x_0-x_1} f(x_0) + \frac{1}{x_1-x_0} f(x_1) + \frac{f^{(2)}(\xi_0)}{2!}(x_0-x_1)$$

let $h = x_1 - x_0$

$$f'(x_0) = -\frac{f(x_0)}{h} + \frac{1}{h} f(x_1) + \frac{f^{(2)}(\xi_0)}{2!}(-h)$$

$$\boxed{f'(x_0) = \underbrace{\frac{f(x_0+h) - f(x_0)}{h}}_{\text{approximation}} - \underbrace{\frac{h}{2} f^{(2)}(\xi_0)}_{\text{Error}}}$$

A similar formula can be obtained for $f'(x_1)$.

(3)

(2) If we consider three points x_0, $x_1 = x_0 + h$, $x_2 = x_0 + 2h$, and rewrite formula (3), we get

$$f'(x_0) = \frac{1}{h}\left[-\frac{3}{2} f(x_0) + 2f(x_1) - \frac{1}{2} f(x_2)\right] + \frac{h^2}{3} f^{(3)}(\xi_0) \quad \text{--- (A)}$$

$$x_0 < \xi_0 < x_0 + 2h$$

$$f'(x_1) = \frac{1}{h}\left[-\frac{1}{2} f(x_0) + \frac{1}{2} f(x_2)\right] - \frac{h^2}{6} f^{3}(\xi_1) \quad \text{--- (B)}$$

$$x_0 < \xi_1 < x_0 + 2h$$

In formula (A), derivative is calculated at x_0 by making use of points on one side of x_0. Since we have used total of three points, it is called Three-Point Endpoint formula, and it can be written in a following more general form.

$$f'(x_0) = \frac{1}{h}\left[\frac{-3}{2} f(x_0) + 2f(x_0 + h) - \frac{1}{2} f(x_0 + 2h)\right] + \frac{h^2}{3} f^{(3)}(\xi_0) \quad \text{--- (C)}$$

$\boxed{x_0,\ x_0 + h,\ x_0 + 2h}$

Here x_0 can be any point in the data & $x_0 + h$, $x_0 + 2h$ are two points on the right side of x_0 i.e, $h > 0$.

However, x_0 lies on the right end of three points such as $\boxed{x_0 - 2h,\ x_0 - h,\ x_0}$ a similar formula can be obtained by replacing h by $-h$ in (C)

$$f'(x_0) = \frac{1}{h}\left[\frac{3}{2} f(x_0) - 2f(x_0 - h) + \frac{1}{2} f(x_0 - 2h)\right] + \frac{h^2}{3} f^{(3)}(\xi_0) \quad \text{--- (D)}$$

Both (C) and (D) are called Three-Point Endpoint formulas

Formula (B) uses ~~bot~~ points from both ends of x_1. This can also be rewritten considering x_0 any general point in the data. $\boxed{x_0-h,\ x_0,\ x_0+h}$

$$f'(x_0) = \frac{1}{h}\left[-\frac{f(x_0-h)}{2} + \frac{1}{2}f(x_0+h)\right] - \frac{h^2}{\cancel{6}}f^{3}(\xi_0)$$

$$x_0-h < \xi_0 < x_0+h$$

This is called Three-Point Midpoint formula.

Becaus of 6 in the denominator, the truncation error in this formula is slightly better than Three-Point Endpoint formula.

~~Example~~

Similarly, if we consider ~~three points~~ five points we obtain five point midpoint & five point endpoint formula.

Five-Point Midpoint Formula

$$f'(x_0) = \frac{1}{12h}\left[f(x_0-2h) - 8f(x_0-h) + 8f(x_0+h) - f(x_0+2h)\right] + \frac{h^4}{30}f^{(5)}(\xi)$$

$$x_0-2h < \xi < x_0+2h$$

Five-Point Endpoint Formula.

$$f'(x_0) = \frac{1}{12h}\left[-25f(x_0) + 48f(x_0+h) - 36f(x_0+2h) + 16f(x_0+3h) - 3f(x_0+4h)\right] + \frac{h^4}{5}f^{(5)}(\xi)$$

Example

Consider the following table.

x	$f(x)$
2.1	-1.709
2.2	-1.3738
2.3	-1.1192
2.4	-0.9160
2.5	-0.7470
2.6	-0.6015

Find all possible derivative formula to find $f'(2.3)$.

Sol
$\Rightarrow$ $h = 0.1$, $x_0 = 2.3$

Three point mid

$$f'(2.3) = \frac{1}{0.1}\left[-\frac{1}{2}f(\underset{x_0}{2.3} - \underset{h}{0.1}) + \frac{1}{2}f(2.3+0.1)\right]$$

$$= \frac{1}{0.1}\left[-\frac{1}{2}f(2.2) + \frac{1}{2}f(2.4)\right]$$

$$= \frac{1}{0.1}\left[-\frac{1}{2}(-1.3738) + \frac{1}{2}(-0.9160)\right]$$

Three point end point $\quad x_0 \quad x_0-h \quad x_0-2h$

2.3, 2.2, 2.1

$$f'(2.3) = \frac{1}{h}\left[\frac{3}{2}f(2.3) - 2f(2.2) + \frac{1}{2}f(2.1)\right]$$

If we consider point on other end of 2.3, ie 2.3, 2.4, 2.5

$$f'(2.3) = \frac{1}{0.1}\left[-\frac{3}{2}f(2.3) + 2f(2.4) - \frac{1}{2}f(2.5)\right]$$

Similary five point mid point can also be used by taking

2.1, 2.2 2.3 2.4, 2.5